

FINAL SEMINAR · OCTOBER 2026

Multi-Page Visually Rich Document Understanding

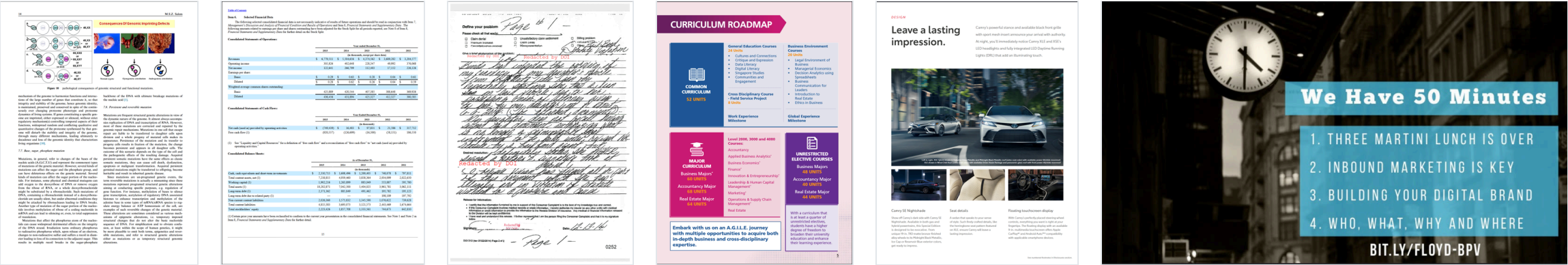
A survey of system design and an empirical attribution of failure

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Visually Rich Document Understanding

A visually rich document is one whose meaning is carried jointly by text, layout, and visual content rather than by text alone. Understanding one is commonly framed as **answering questions over it**.



Representative pages from MMLongBench-Doc

- **Text.** Small, typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.
- **Vision.** Figures, charts, photographs, and handwriting carry meaning that no text extraction recovers.
- **Layout.** Position is semantic, and reading order must be recovered rather than assumed.
- “Which year had the largest gap between projected and actual spending?”
- “How many of the diagrams in this manual carry a safety warning?”
- “What does the appendix table referenced in Section 3.2 report for the second condition?”
- “What was the total headcount at the end of the year?”

General-Domain MLLMs and VRDU

VRDU long predates general-domain MLLMs; what has changed is that those models now sit at the centre of it. Closed-source assistants have steadily absorbed VRDU techniques into their products, while open-weight models carry the same techniques to documents that cannot leave the premises.

Closed-source products

Three routes cover most of what a consumer assistant does with an uploaded document.

- **Read it whole.** Parse the PDF for text, render it to per-page images, and place both in the context window.
- **Read it on demand.** The file stays on disk, and tool calls extract or search particular pages.
- **Retrieve over it.** Content beyond the context window is indexed and searched rather than ingested.

ChatGPT · Claude · Gemini · NotebookLM

Open-weight models, on premises

Much document work involves material that cannot be sent to a third party at all.

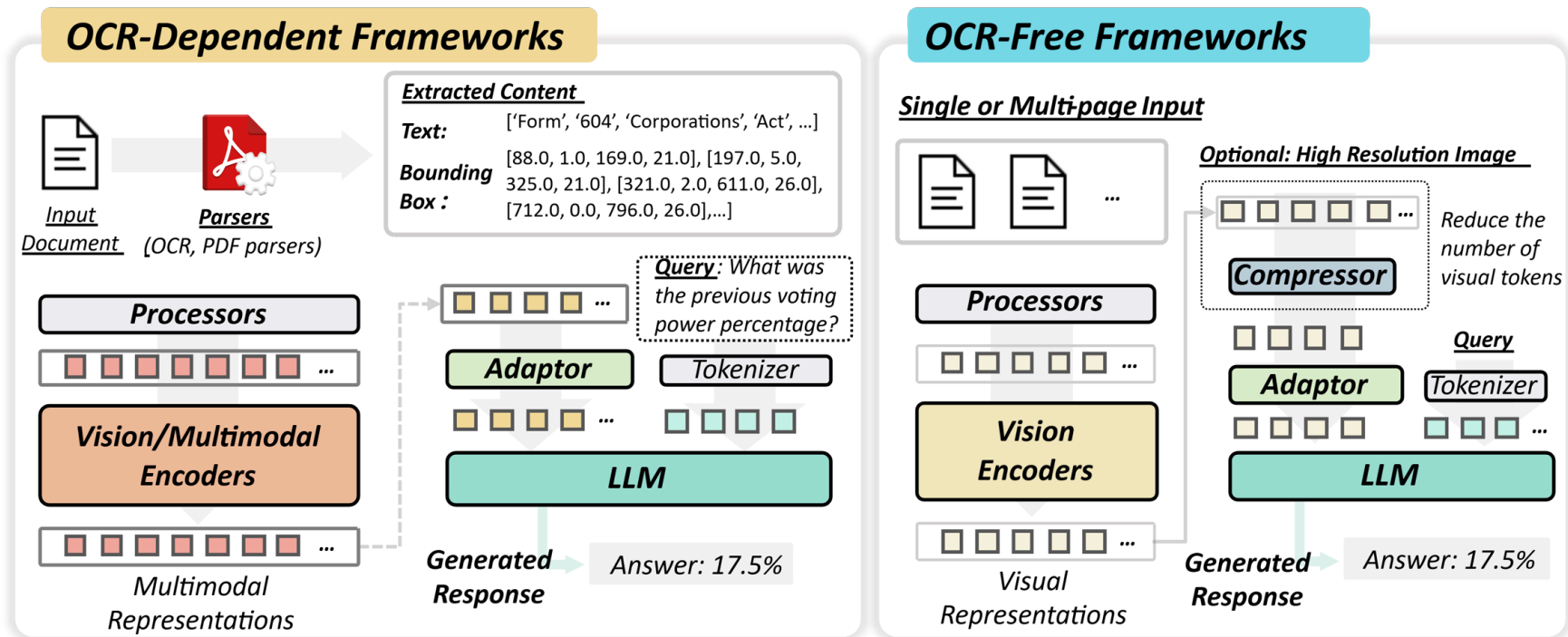
- **Documents that cannot leave.** Medical records, legal and financial filings, and mining and resources data.
- **The same pipelines, local weights.** Open-weight backbones rebuilt behind the firewall rather than called over an API.
- **Hardware is the binding constraint.** Quantisation, retrieval, and page selection do the work the context window cannot.

Qwen3-VL · Gemma · InternVL

Both are document-understanding pipelines. The frontier is how these backbones can be leveraged, together with the machinery built around them, to answer questions over documents they **cannot process directly**.

Single-Page VRDU Overview

The single-page setting is largely solved, and multi-page systems inherit their per-page machinery from it.



OCR-dependent and OCR-free architectures for MLLM-based VRDU. Adapted from Ding et al.

OCR-dependent

Parse the page first, recovering text with its bounding-box coordinates. Captures structure without heavy visual pretraining, but ties accuracy to the parser's.

OCR-free

A vision encoder reads the rendered page from pixels. Handles handwriting and irregular layouts, but needs high resolution, so compression usually follows.

Challenges of the Multi-Page Setting

- **The document exceeds the context window.** Long multi-page documents routinely outrun what a backbone can hold, so the system must select which parts of the document reach the MLLM at all.
- **Scaling the context does not resolve it.** Long-context models attend poorly to evidence buried mid-input, and memory and compute grow with sequence length, so the achievable context is bounded by hardware.
- **Every required evidence unit must be located.** Answering often means integrating evidence from several parts of the document at once, and missing any one of them is enough to answer incorrectly or not at all.
- **Cross-page structure is not recoverable in isolation.** Section hierarchy and cross-references cannot be read off any single part of the document, yet questions often need them to gather the evidence in the first place.

Seminar Overview

Two papers came out of this Honours project. The rest of the seminar follows them in turn.

1 **Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding**

Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

Main Conference of EMNLP 2026

NEXT MP-VRDU organised through evidence management: a taxonomy of four architectural families, multimodal representation, and inference-time and training strategies.

2 **Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution**

Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

Submitted to EACL, under review

THEN Failure attributed to three loci, representation, selection, and reasoning, through controlled interventions on a single pipeline.

Evidence Management

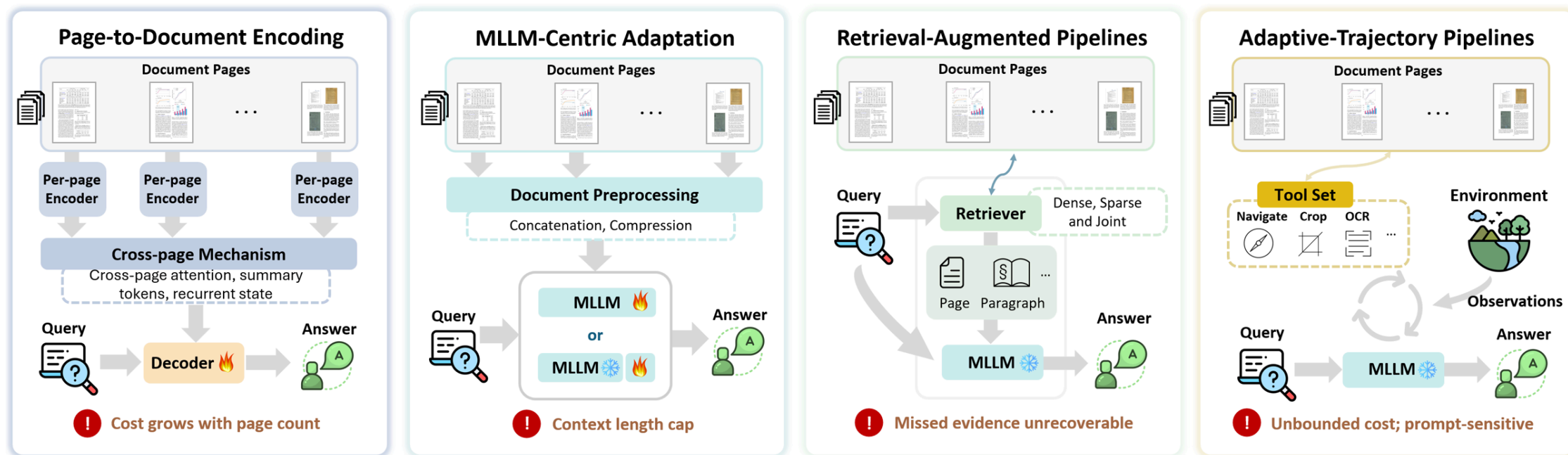
The defining problem in MP-VRDU is **evidence management**. Performance depends not only on the capability of the backbone, but on how evidence is acquired, preserved, prioritised, and used across the whole pipeline.

Acquired	Preserved	Prioritised	Used
Turning the document into something the later stages can read at all.	Deciding what survives a fixed token budget once the document will not fit.	Choosing which of what survives is actually placed before the reasoner.	Reasoning over what arrives, and going back for more when it is not enough.
page rendering	per-page summary tokens	similarity retrieval	prompted reasoning
OCR and PDF parsers	visual-token compression	structural navigation	iterative re-retrieval
element cropping	recurrent memory state	reranking	tool calls
render resolution	sparse cross-page attention	adjacent-page expansion	self-refinement
text-layer extraction	structure-aware chunking	graph traversal	agentic decomposition

Every architecture in this section is a different answer to the same four questions. **What a system can do is fixed by where it makes these decisions.**

Architectural Families

We organise the field by architecture rather than by OCR dependency, because the architecture is what fixes how cross-page evidence is selected, preserved, and exposed for reasoning.



Snowflake indicates typically frozen, fire indicates typically trained

Read left to right, each family pushes evidence handling further out of the backbone: first compression, then retrieval, then iterative orchestration. **None of them removes the bottleneck; each relocates it**, and the red note under every panel is that family's price.

Multi-Page Multimodal Representation

Every evidence decision is enacted over a modality, and its cost depends on how that modality is represented. Each one trades **page coverage** against **reading fidelity**.

Text

- **Volume.** Text across pages routinely exceeds what the backbone can process, so it is either compressed into a fixed budget before the question is seen, or indexed and retrieved.
- **Continuity.** Paragraphs, tables, and section context break at page boundaries, and every repair pulls in neighbours by proximity rather than by question.

Visual

- **Resolution.** Token cost scales with page count times per-page resolution, forcing a choice between compressing every page and reading a retrieved subset at full fidelity.
- **Signal competition.** One visual budget must carry text-bearing pixels, figures and charts, and layout cues within the same page.

Cross-page structure

- **Hierarchy and reading order.** Sections, lists, and captions span page boundaries, and parser errors propagate into every downstream retrieval step.
- **Cross-page references.** “See Figure 3” has to be resolved to its referent on another page, which similarity scoring cannot do.

Heterogeneity cuts across all three and receives the least attention: **text density and visual complexity vary sharply from page to page**, yet almost every system applies one fixed budget to the whole document.

Retrieval and Navigation

Retrieval narrows a long document to compact evidence units. Two families have emerged, distinguished by whether passages are **scored independently** against the query or **navigated through document structure**.

INFERENCE-TIME

- **Similarity-based.** Score each passage against the query independently and return the top-ranked subset. Dense retrieval has consolidated around late-interaction encoders such as ColPali, which align query tokens to page patches by MaxSim; BM25 persists as a sparse baseline, and a second LLM reranking stage commonly refines the recall.
- **Relation-aware.** Retrieve along explicit structure instead. Tree variants group passages under recovered section headings so retrieved chunks keep their parent context; graph variants make headings, entities, and cross-references nodes with typed edges, so a controller can traverse to related passages rather than rely on similarity.

TRAINED

- **Modality alignment.** Learn visual page embeddings that stay aligned with OCR-derived text, so page images can be retrieved without late interaction's storage and latency cost.
- **Cross-page fusion.** Enrich each page representation with neighbouring-page context through a lightweight LoRA head, making evidence spread across adjacent pages retrievable as one unit.
- **Logical relevance.** Train graph scorers to rank pages by their role in cross-page reasoning paths, turning retrieval into a learned multi-hop walk rather than a single similarity match.

Independent scoring handles paraphrase and cross-modal queries well, but **cannot follow a reference or a multi-hop dependency that no single passage carries**.

Reasoning Strategies

Reasoning acts over a **single fixed evidence buffer** assembled before the call. The step itself acquires nothing new, so its contribution lies in how faithfully the delivered evidence is used, not in recovering evidence the buffer omits.

INFERENCE-TIME

- **Sequential.** Chain-of-thought elicits intermediate reasoning before the answer, which binds evidence from different pages step by step, decomposes multi-hop questions into single-hop ones, and externalises the trace for verification. Self-reflection extends the same thread with iterative revision against the buffer, optionally under a verifier.
- **Parallel.** Self-consistency samples several chains and takes the majority, or has an LLM adjudicate between reasoning paths. Tree-of-thoughts explores branches with backtracking, but has seen limited uptake, since in MP-VRDU evidence sparsity rather than search branching is the dominant difficulty.

TRAINED

- **Trace imitation.** Supervise the backbone on demonstration chains wrapped in fixed schema tags such as `<think>`, `<evidence>`, and `<answer>`, refined with DPO to prefer faithful chains over hallucinated shortcuts.
- **Reward-driven.** Optimise the reasoning policy with GRPO against a composite reward of answer accuracy, evidence-page recall, and format compliance. Usually warm-started from imitation rather than learned from scratch, so the two are sequential rather than alternative.
- **Uncertainty calibration.** Train an aggregation head that selects across candidate outputs by calibrated uncertainty, giving a learned self-consistency signal without sampling several chains at test time.

All of it is independent of architectural family, and all of it is bounded the same way: **missing evidence cannot be reasoned back into existence.**

Agentic Methods

The current frontier. A controller selects each evidence-gathering action **at inference time** rather than executing a plan fixed in advance, so compute scales with question difficulty rather than document length, and the action sequence becomes an inspectable audit trail.

INFERENCE-TIME

- **ReAct and tool use.** Thought and action interleave in one stream, each observation feeding the next step. Actions are either document-side tool calls to retrievers, parsers, and OCR engines, or structured navigation over a pre-built representation, which trades dependence on tool fidelity for dependence on the index.
- **Agentic decomposition.** Split the workflow across controllers, each on a narrower input than one backbone doing everything. Modality-specialised variants read text and vision separately and reconcile; role-specialised variants assign plan, execute, verify, and answer, with routing back when a stage detects an error.

TRAINED

- **Trajectory distillation.** Clone a stronger teacher's trajectories by supervised fine-tuning on think-action-observation turns, with rollouts collected in closed-loop interaction and filtered for format validity, answer correctness, and evidence coverage.
- **Reinforcement-learned.** Optimise the policy directly against trajectory-level rewards combining answer correctness, evidence recall, and tool-use shape, over a constrained action space. Learned termination is a recurring concern, with entropy-based stopping letting the model commit once confidence stabilises.

The price of that flexibility: inference cost is hard to bound in advance, controllers are prompt-sensitive and can commit early to the wrong evidence, and the system still **inherits the recall ceiling of every tool it consults**.